PROJECT SPARK | LEVEL 16

Series LED Current Challenge - Unity creation guide

Design: a single 5 V series loop with a replaceable 100 ohm resistor, red LED, and green LED. The starter resistor is 330 ohm and must be exchanged while power is off.

1 | Correct topology

```
+5V -- Switch -- 100R -- Red LED -- Green LED -- GND
```

Red cathode connects directly to green anode. This level has **one current path**; no 470 ohm parallel branch.

2 | Parts and prefab assembly

Scene part	Setup
PowerSupply_01	Nominal 5 V; Battery_positive and Battery_negative terminal IDs
Power_Switch	Series power switch, output to replaceable resistor
Workstation resistor	Register_Slot_2; initial 330 ohm, correct shelf part 100 ohm
Red LED	Fixed SparkLED component ID led, anode LED_Anode, cathode LED_Cathode
Green LED	Fixed SparkLED component ID led_2, anode led_2_Anode, cathode led_2_Cathode
Meter (optional)	Measure voltage across installed resistor: R_LIMIT_Input to R_LIMIT_Output

3 | Connect wires in exact order

#	From	To
1	Battery_positive	Power_Switch input
2	Power_Switch output	R_LIMIT_Input
3	R_LIMIT_Output	LED_Anode
4	LED_Cathode	led_2_Anode
5	led_2_Cathode	Battery_negative
6 optional	LCD (+)	R_LIMIT_Input
7 optional	LCD (-)	R_LIMIT_Output

LEVEL 16 | UNITY BINDING CHECKLIST

4 | Universal workstation bridge

Setting	Value
Slot ID	Register_Slot_2
Component binding ID	R_LIMIT_Component
Input terminal binding ID	R_LIMIT_Input
Output terminal binding ID	R_LIMIT_Output
Electrical type	SparkResistor
Terminal suffixes	_Input and _Output

5 | Primary SparkLevelSceneBindings

Runtime component ID: R_LIMIT_Component
Runtime terminal IDs: R_LIMIT_Input, R_LIMIT_Output
Fixed electrical component IDs: led, led_2, PowerSupply_01
Fixed terminal IDs: LED_Anode, LED_Cathode, led_2_Anode, led_2_Cathode, Battery_positive, Battery_negative.

These IDs must be added to the ORIGINAL Level_16_Layout prefab, not the Play Mode clone. Do not add runtime aliases to fixed component arrays.

6 | Import the definition

1. Back up the original Level_16.asset. 2. Select the ORIGINAL asset at Assets/My_Assets/_Project_Spark/Prefabs/Levels/Level_16_Layout/Level_16.asset. 3. Import Level_16_SeriesLED_Import.txt with the existing Project Spark TXT importer. 4. Verify levelId = level_16 and Required Target Count = 5. 5. Save asset and layout prefab.

7 | Five completion targets

Target	Type	Requirement
Red LED conducting	ComponentConducting	led is conducting
Green LED conducting	ComponentConducting	led_2 is conducting
Resistor series current	CurrentRange	R_LIMIT_Component: 8-12 mA
Resistor voltage	VoltageRange	R_LIMIT_Input to R_LIMIT_Output: 0.8-1.2 V
Supply current	CurrentRange	PowerSupply_01: 8-12 mA

LEVEL 16 | VALIDATION & TESTING

8 | Expected electrical response

Nominal model: 5 V - (2 V red LED + 2 V green LED) = 1 V across 100 ohm, so the series current is about 10 mA. Both LEDs must carry the same current, and source current must match the series current. The 330 ohm starter resistor would give roughly 3 mA with this simplified model.

Important: these are design assumptions, not confirmed readings from your SparkLED solver. Measure your actual circuit audit after installing 100 ohm. Adjust the range targets if the modeled green LED drop differs.

9 | In-Editor creation sequence

1. Duplicate/prepare Level_16_Layout and configure Level 16 definition reference.

2. Remove the Level 15 parallel resistor branch, retaining 5 V supply and switch.

3. Arrange fixed red and green SparkLED components in a single series chain.

4. Configure workstation slot Register_Slot_2; starter 330 ohm, correct 100 ohm on shelf.

5. Add three runtime IDs and bridge rule to the original layout prefab.

6. Bind both fixed LEDs to their real SparkLED components (not GameObject references).

7. Import TXT, save asset and prefab; restart Play Mode.

8. Wire the series loop; install 100 ohm with power off; close switch.

9. Run solved source current audit; compare R_LIMIT, both LEDs, and source current.

10. Check [SPARK TARGET] logs for five Passed=True targets.

10 | Troubleshooting

Symptom	First check
LED target fails	Fixed ID led / led_2 resolves to actual SparkLED and terminal polarity
Resistor current target fails	Bridge resolves R_LIMIT_Component, R_LIMIT_Input, R_LIMIT_Output on current clone
Resistor voltage target fails	Terminal IDs resolve and 0.8-1.2 V band matches actual model
Source target fails	Series path closed; verify no additional parallel load remains
Target values report zero	Check resolved IDs / bindings rather than modifying electrical solver

Asset identity retained: c6e74415442d6994ea9cdc11fd6168fb. The package intentionally does not change prefabs or runtime source code.